De *diye*

Deniz Özyıldız
Universität Stuttgart

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De *diye*

tl;dr

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  - Travis: yeah.
  - Faruk: no.
  - Deniz: maybe?

Two types of clauses in Turkish

-DIK nominalizations and *diye* clauses

Similarities

Turkish -DIK nominalizations and *diye* clauses…

- ✧ have a different internal morphosyntax.
- ✧ have overlapping syntactic distributions.
- ✧ sometimes give rise to similar truth-conditions.

(1) [Kar yağ-ıyor diye] düşün-dü.
snow fall-PRES DIYE think-PST
She thought that it was snowing.

(3) [Kar yağ-dığ-ın-ı] düşün-dü.
snow fall-NMZ-3S.POSS-ACC think-PST
She thought that it was snowing. not: 'think *about* our arrival'

But their distribution is not identical and they often give rise to different interpretive effects.

- **Indexical shift**
  Possible in *diye* clauses, impossible in nominalizations.

  (4) Molly$_1$ said that I$_1$ was happy. (pseudo-English)

- **Factivity** know p $\rightsquigarrow$ p
  The truth of a *diye* clause is never$^?$ entailed.
  With nominalizations, this depends on the verb.

- **Linguistic production**
  Diye clauses sometimes give rise to the inference that their content was uttered, thought, entertained, etc.

-DIK nominalizations and *diye* clauses

Some semantic differences

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  Diye clauses sometimes give rise to the inference that their content was uttered, thought, entertained, etc.

Today

What do these dfferences tell us about the syntax and semantics of *diye* clauses?

“The reason we care is that languages care. Languages to special things with [complement clauses].” Constant (2014)

The linguistic production inference

The linguistic production inference is available with *diye*

(5) a. Tunç [ne oldu? diye] şaşırdı.
Tunç what happened DIYE got surprised
Tunç got surprised, saying "What happened?"

b. Dilara da öyle dedi.
Dilara too so said
Dilara said that too.

$\rightsquigarrow$ Tunç and Dilara both said "What happened?"

The linguistic production inference is not available with DIK

(6) a. Tunç [ne olduğuna] şaşırdı.
Tunç what happen.NMZ got surprised
Tunç got surprised by what happened.

b. #Dilara da öyle dedi.
Dilara too so said
Dilara said that too.

$\not\rightsquigarrow$ Tunç said "What happened?"

The linguistic production is obligatory when available

(7) #Bebek/kedi [ne oldu? diye] şaşırdı.
baby/cat what happened DIYE got surprised
#The baby/cat got surprised, saying “What happened?”

(8) Bebek/kedi [ne olduğuna] şaşırdı.
baby/cat what happen.NMZ got surprised
The baby/cat got surprised by what happened.

An additional contrast: *Deny* something vs. *deny* by saying something

(9) (Birşeyi) [hile yaptı diye] inkâr etti.
something cheat did DIYE denied
He denied something, saying that he cheated.

(odd if we can't recover a direct object)

(10) [Hile yap-tığ-ın-ı] inkâr etti.
cheat do-NMZ-3S-ACC deny
He denied that he cheated.

⇝ He said he didn't cheat.

The inference is not always as clearly triggered

Do we get a similar inference in (11)?

Yes, possibly.

(1) [Kar yağ-ıyor diye] düşün-dü.
snow fall-PRES DIYE think-PST
She thought that it was snowing.

Not really?

(11) [Kar yağ-ıyor diye] bil-iyor.
snow fall-PRES DIYE BIL-PRES
She believes that it's snowing.

Naturally occurring *diye inan-*

No linguistic inference

(12)

[Erdoğan	datayı	saklıyor	diye]	inanıyor	ve	bunu
Erdoğan	the data	is hiding	DIYE	believe	and	this

kanıtlamaya	çalışıyorsanız, …
to prove	you're trying

If you believe that Erdoğan is hiding the data and you're trying to prove this, …

Upshot

- ✧ The linguistic production inference is sometimes triggered, but not always (with *bil-*, *inan-*).
- ✧ When triggered (with ‘open the door,’ ‘deny’),
  - · it is obligatory,
  - · it can clearly be attributed to *diye*.

Different composition strategies for -DIK nominalizations and *diye* clauses ○○●○

Two guiding hypotheses

partially independent from one another

- ✧ Syntax
  - Nominalizations are **arguments**.
  - *Diye* clauses are **adjuncts**. (to, e.g., vP or VP)
- ✧ Semantics
  - *Diye* clauses mean **"to say, and"**.
  - Nominalizations get their meaning from their function.

$$[NMZ\ V]_{VP} \rightsquigarrow [\![V]\!]([\![NMZ]\!])$$

These hypotheses go a long way.

- But… are *diye* clauses always adjuncts, or only sometimes?
- How to account for the observation that they only sometimes trigger a linguistic production inference?

Composition with (derived) intransitive predicates

(2) [Kar yağ-ıyor diye] zıpladı.
snow fall-PRES DIYE jump up-PST
She jumped up, said that it was snowing.

(13) *[Kar yağdığını] zıpladı.
snow fall.NMZ jump up-PST
Intended: She jumped up, said that it was snowing.

This suggests that:

- *diye* clauses can be adjuncts,
- DIK clauses can't be adjuncts/have to be arguments.

But, can we make a stronger claim for *diye* clauses?

Argument- vs. adjuncthood

The ‘case on the causee’ test
(Çetinoğlu & Butt 2008; Yıldırım-Gündoğdu 2017, 2018; Akkuş 2021)

(14) Core contrast

a. Travis Jack’i giy-dir-di.
Travis Jack.ACC wear-CAUS-PST.3S
Travis dressed Jack (lit. made him wear).

b. Travis Jack’e giysi(-yi) giy-dir-di.
Travis Jack.DAT clothes-ACC wear-CAUS-PST.3S
Travis made Jack wear clothes. …*Jack’i…

Argument- vs. adjuncthood

The 'case on the causee' test—spot checks
(Çetinoğlu & Butt 2008; Yıldırım-Gündoğdu 2017, 2018; Akkuş 2021)

Adjuncts don't trigger DAT, null objects can.

(15) a. Travis Jack'i hızlıca/bugün giy-dir-di.
Travis Jack.ACC quickly/today wear-CAUS-PST.3S
Travis dressed Jack quickly/today.

b. Travis Jack'e ∅ giy-dir-di.
Travis Jack.DAT wear-CAUS-PST.3S
Travis made Jack wear it.
(To answer, e.g., "What happened to my hat?")

Argument- vs. adjuncthood

The 'case on the causee' test applied to nominalizations

Accusative nominalizations obligatorily trigger dative causees.

(16) Bana [var-dığ-ın-ı] düşün-dür-dü.
1S.DAT arrive-NMZ-2S-ACC think-CAUS-PST.3S
She made me think that you arrived.

(17) *Beni [var-dığ-ın-ı] düşün-dür-dü.
1S.ACC arrive-NMZ-2S-ACC think-CAUS-PST.3S
Int. She made me think that you arrived.

⇒ Nominalizations pattern like other arguments in the language, which should ideally be reflected in the semantic composition.

Argument- vs. adjuncthood

The 'case on the causee' test applied to *diye* clauses

Diye clauses are compatible with accusative causees.

(18) Beni [var-dın diye] düşün-dür-dü.
1S.ACC arrive-PST.2S DIYE think-CAUS-PST.3S
She made me think that you arrived.

⇒ This suggests the **availability of an adjunct parse**.

Argument- vs. adjuncthood

The 'case on the causee' test applied to *diye* clauses

They are also compatible with dative causees.

(13) Bana [var-dın diye] düşün-dür-dü.
1S.DAT arrive-PST.2S DIYE think-CAUS-PST.3S
She made me think that you arrived.

⇒ This suggests the **availability of an argument parse**.

(The exact conditions of when/for who this is possible are unclear.)

Naturally occurring ACC and DAT causees

(14) Cesedini parçalayıp lögara atmak lazım mezarı bile olmasın **diye düşündürdü** **beni**.

That made me (ACC) think that they shouldn't even have a grave.

(15) Belki de psikoloji okumasının karakter çözümlemesinde çok büyük faydası var **diye düşündürdü** **bana**.

That made me (DAT) think that their having studied psychology has a great effect on their character analyses.

This same test can be run in French as well, and other languages with the relevant ACC/DAT alternation.

(16)

Je	lui	ai	fait	{dire,	penser}	que	j'étais arrivé.
I	3S.DAT	have	made	say	think	that	I=was arrived

I made him {say, think} that I had arrived.

(17)

*Je	l'ai	fait	{dire,	penser}	que…
I	3S.ACC=have	made	say	think	that

⇒ *Que* clauses pattern like arguments.

An argument parse?

A natural conclusion is that *diye* clauses are sometimes arguments.

But they would then have to be a very special kind of argument.

- Restricted from subject positions.
- And from object positions with certain predicates.

Diye clauses can't be subjects

We've seen them in object position.

But they are restricted in subject position:

(18) *[Var-dın diye] {doğru, belli, kesin}.
arrive-PST.2S DIYE true obvious certain.COP.3S
Int. It is {true, obvious, certain} that you arrived.

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Int. It is {true, obvious, certain} that you arrived.

Which raises questions about their syntactic position with passivized attitude predicates:

(19) [Var-dın diye] düşün-ül-dü.
arrive-PST.2S DIYE think-PASS-PST.3S
It was thought that you arrived.

⇒ Q&A. Many thanks to Faruk Akkuş (2021, p.c.) for discussion.

Diye clauses can't always be objects

Interpretive differences with nominalizations

(20) [Gel-diğ-in-i] inkâr et-ti.
come-NMZ-2S-ACC deny-PST.3S
She denied that you came.

(21) [Gel-din diye] inkâr et-ti.
come-PST.2S DIYE deny-PST.3S
She denied something, saying that you came.

This difference is hard to capture if we assume that *diye* is the argument of "deny."

Diye clauses can't always be objects

Interpretive differences with nominalizations

(22) [Ne ol-duğ-un-u] cevapla-dı.
what be-NMZ-3S-ACC answer-PST.3S
She answered the question of what happened.

(23) [Ne ol-du diye] cevapla-dı.
what be-PST.3S DIYE answer-PST.3S
She answered a question, saying "What happened?"

This difference is hard to capture if we assume that *diye* is the argument of "answer."

Diye clauses can't always be objects

Interpretive differences with nominalizations

(24) [Ne ol-duğ-un-u] tekrarla-dı.
what be-NMZ-3S-ACC repeat-PST.3S
She repeated what happened.

(25) [Ne ol-du diye] tekrarla-dı.
what be-PST.3S DIYE repeat-PST.3S
She repeated the question "What happened?"

This difference is hard to capture if we assume that *diye* is the argument of "repeat."

Upshot

- Nominalized clauses pattern like arguments.
- Some *diye* clauses pattern like modifiers and some like arguments.
- But we can't equate DIK nominalizations and *diye* clauses' argument parses.

  That leads to wrong expectations about their distribution and interpretation.

λ

What would it take to provide a syntax and semantics for nominalizations and *diye* clauses, if we assume that *diye* clauses are always adjuncts?

(Pending a possibly better analysis of argument-like cases.)

The Özyıldız & Uegaki proposal

Clauses can be modifiers, or arguments: Evidence from alternations in factivity and answer-orientedness in Turkish and Japanese

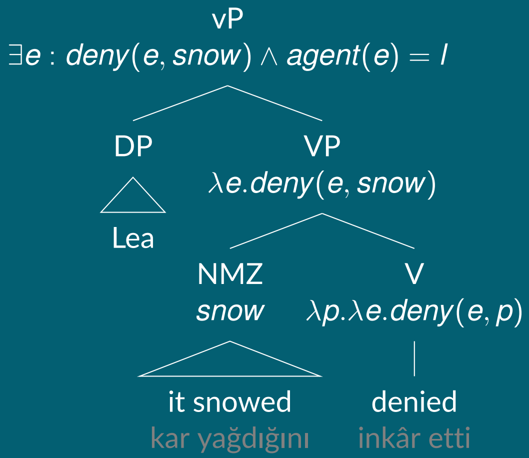
Deniz Özyıldız & Wataru Uegaki
Universität Konstanz, University of Edinburgh

June 7, 2024

Abstract

Turkish and Japanese attitude predicates combine with two main kinds of embedded clauses: Nominalizations, and clauses introduced by the morphemes *diye* (Turkish) and *to* (Japanese). It has been previously been observed that the choice of the embedded clause conditions the availability of a factive inference, for declaratives under certain predicates (Özyıldız 2017; Kusumoto 2017). We describe the syntactic and semantic properties of interrogative nominalizations and *diye*/*to*-clauses. We show that the interrogative nominalizations give rise to familiar answer-oriented inferences with responsive predicates, but that interrogative *diye*/*to*-clauses must remain question-oriented. In particular, they give rise to the entailment that the attitude holder linguistically produce the question that they introduce. We propose a compositional fragment where attitude predicates take nominalizations as arguments, which they may impose semantic restrictions on, and where *diye*/*to*-clauses modify and enrich attitude meanings with a linguistic production inference. We end with a discussion of whether this proposal is viable uniformly for all instances of *diye*/*to*-clauses.

Nominalized clauses



Lexical knowledge:
$[\exists e : deny(e, snow) \wedge agent(e) = l] \Rightarrow$ Lea said it didn't snow

Predicates may impose semantic restrictions on argument NMZs.

Diye clauses

Diye clauses add a *say* ing event to the matrix event $V(e_2)$.

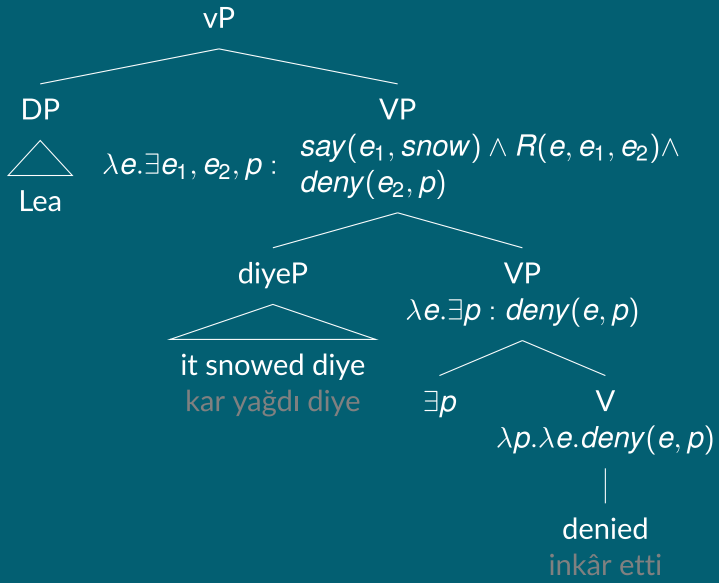
(26) $[\![S\ diye]\!] =$
$$\lambda V_{vt}.\lambda e.\exists e_1, e_2 : say(e_1, S) \wedge R(e, e_1, e_2) \wedge V(e_2)$$

The predicate *say* varies between contentful and bleached.

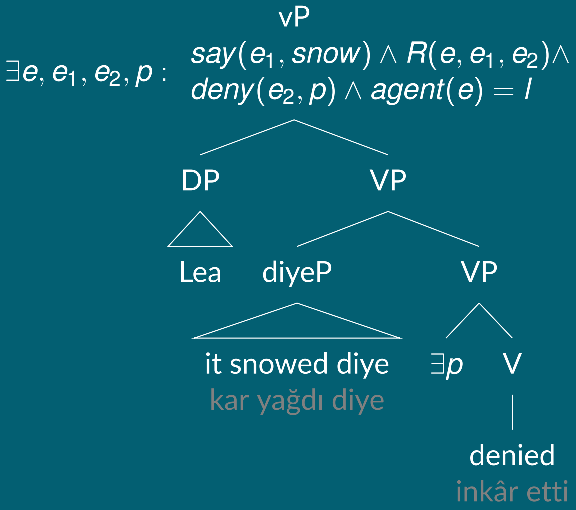
(27) This rumor says that it's snowing. (T. Major, p.c.)

- ✧ S is saturated by the clause.
- ✧ V is saturated by the main event predicate ('think,' 'deny,' 'jump up').
- ✧ R relates the *say* ing event e_1, the matrix event e_2, and a 'container' event.

Diye clauses



Diye clauses



How to resolve R

In some cases, let $R := \lambda e.\lambda e'.\lambda e''.e = e' = e''$

Then, (28) simplifies into (29):

(28) $\exists e, e_1, e_2, p : \begin{array}{l} say(e_1, snow) \wedge R(e, e_1, e_2) \wedge \\ deny(e_2, p) \wedge agent(e) = l \end{array}$

(29) $\exists e, p : \; say(e, snow) \wedge deny(e, p) \wedge agent(e) = l$

This entails that Lea said that it snowed and that she denied something in the same event.

How to resolve R

In some cases, let $R := \lambda e.\lambda e'.\lambda e''.e = e \oplus e'$

(merelogical sum)

This is useful for *diye* clauses combining with non-attitude predicates.

(30) Lea [kim geldi diye] zıpladı.
Lea who arrived DIYE jumped up
Lea jumped saying/wondering: "Who came?"

(31) $\exists e, e_1, e_2 : \begin{array}{l} \mathit{say}(e_1, \text{who came?}) \wedge \\ e = e_1 \oplus e_2 \wedge \\ \mathit{jump\ up}(e_2) \wedge \\ \mathit{agent}(e) = l \end{array}$

This entails that Lea is the agent of a complex event, one part jumping up, another part saying "Who came?"

Accounting for the argument-like uses of *diye* clauses

The argument-like behavior of some *diye* clauses is explained by optional silent internal arguments in the VP.

(13′)	Bana	[var-dın	diye]	$\varnothing$	düşün-dür-dü.
	1S.DAT	arrive-PST.2S	DIYE		think-CAUS-PST.3S

She made me think that you arrived.

The disappearance of the linguistic production inference is explained by assuming an *R* of identity and a bleached *say*.

(32) $\exists e, p: \quad say(e, snow) \wedge think(e, p) \wedge agent(e) = I$

Concluding remarks

There are languages with complement clauses that behave differently from each other, syntax and semantics-wise.

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There are ready-made solutions: Pure complementation (the textbook/Hintikka), pure modification (Elliott), mixtures of various kinds (Kratzer, a.o.)

Concluding remarks

There are languages with complement clauses that behave differently from each other, syntax and semantics-wise.

There are ready-made solutions: Pure complementation (the textbook/Hintikka), pure modification (Elliott), mixtures of various kinds (Kratzer, a.o.)

There is much to sort through. But what we can be sure of is:

- We need to admit multiple paths to composition.
- We need to think about the possibility that some classes of clauses have differently patterning subclasses? (Adjunct vs. argument-like *diye* clauses.)
  And how to model them in an empirically satisfying way.

Appendix

A non-exhaustive list

The work from today draws on ongoing work with Wataru Uegaki, Faruk Akkuş & Travis Major.

It is made better by audiences in Toronto, New Orleans, Konstanz, Edinburgh, Tübingen, Stuttgart, Vienna, and USC (!), and by conversations with Tanya Bondarenko, Ellen Brandner, David Diem, Kajsa Djärv, Richard Faure, Miray Gökkaya, Ali Doruk Işıldak, Jérémy Pasquereau, Maribel Romero & Marc-Matthias Zymla.

Indexical shift

(33) Tunç [annem Ankara'da diye] düşündü.
Tunç my mom in Ankara DIYE thought
Tunç thought that {his, my} mom was in Ankara.

(34) Tunç [annemin Ankara'da olduğunu] düşündü.
Tunç my mom in Ankara be.NMZ thought
Tunç thought that {#his, my} mom was in Ankara.

Factivity alternations

(35) Tunç [kar yağıyor diye] biliyor.
Tunç snow fall DIYE BIL
Tunç thinks that it's snowing.

(36) Tunç [kar yağdığını] biliyor.
Tunç snow fall.NMZ BIL
Tunç knows that it's snowing.

Are *diye* clauses islands?

Baseline: No adjunct extraction out of a relative clause.

(37) *Tunç [niçin gelen kuryeyi] azarladı?

Intended: What is the reason s.t. Tunç scolded the delivery person who came for that reason?

Diye clause + manner of speech verb: Extraction OK?

(38) Tunç [kurye niçin geldi diye] bağırmıştı?

What is the reason s.t. Tunç screamed that the delivery person came for that reason?

Diye clause + non-attitude VP: Extraction OK?

(39) Tunç [kurye niçin geldi diye] kapıya koştu?

What is the reason s.t. Tunç ran to the door, saying that the delivery person came for that reason?

Are *diye* clauses islands?

Reasoning:

- ✧ In (39), we can be sure? that the *diye* clause is an adjunct.
- ✧ If extracton is possible, *diye* clauses are not reliably islands even when forced to be adjuncts.
- ✧ And we can't conclude based on this test about whether the *diye* clause in (39) is an adjunct or not.

A more certain consequence of the subjecthood facts

(18) *[Var-dın diye] {doğru, belli, kesin}.
arrive-PST.2S DIYE true obvious certain.COP.3S
Int. It is {true, obvious, certain} that you arrived.

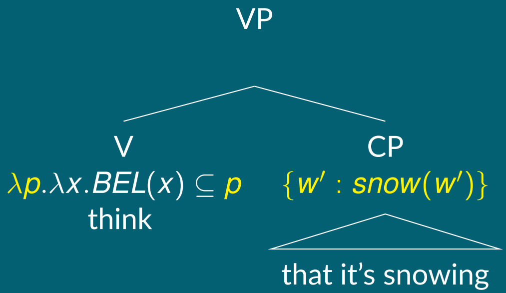
The contrast between unacceptable subject *diye* clauses and (40) suggests that these can't *freely* be introduced by silent (pro)nouns.

(40) [[Var-dın diye] bir bilgi] doğru.
arrive-PST.2S DIYE one information true.COP.3S
A piece of information that says that you arrived is true.

The complementation strategy (Hintikka + textbook semantics)

Two (among many) competing answers

- ✧ Clauses are syntactic and semantic complements of verbs.
- ✧ Attitude reports' interpretive properties are mostly decided by verbs.



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VP

$\lambda x.BEL(x, w) \subseteq \{w' : snow(w')\}$

V

$\lambda p.\lambda x.BEL(x) \subseteq p$

think

CP

$\{w' : snow(w')\}$

that it's snowing

The modification strategy — Elliott (2017), a.o.

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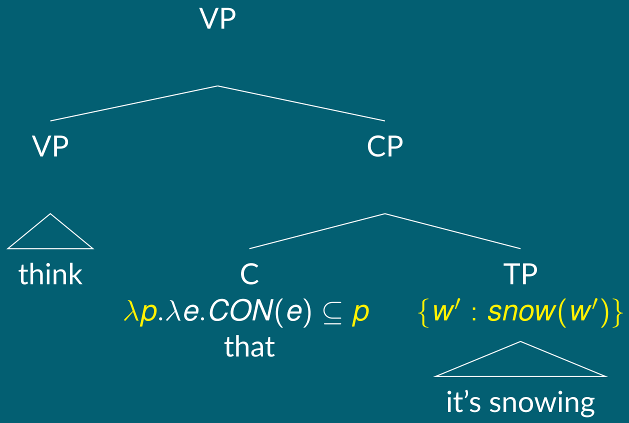

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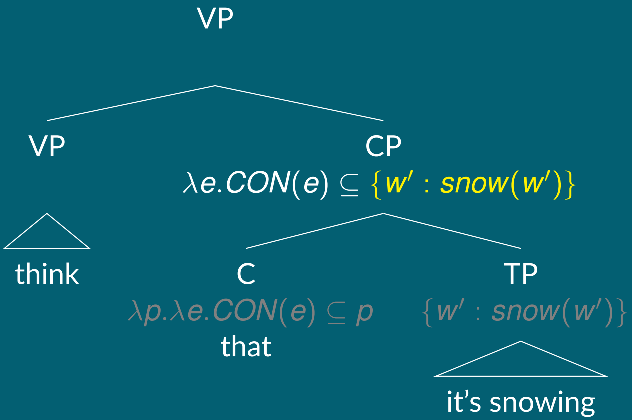

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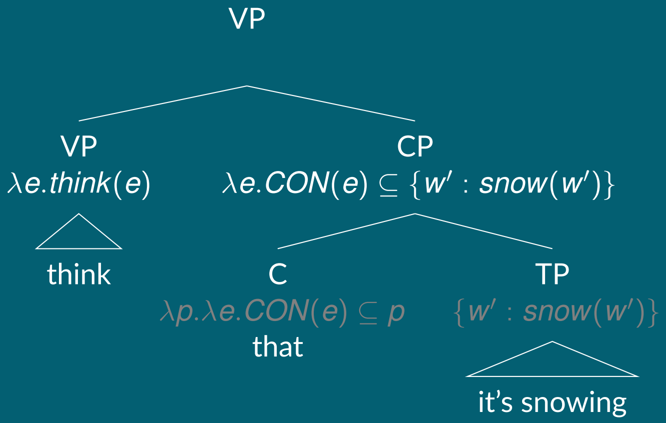

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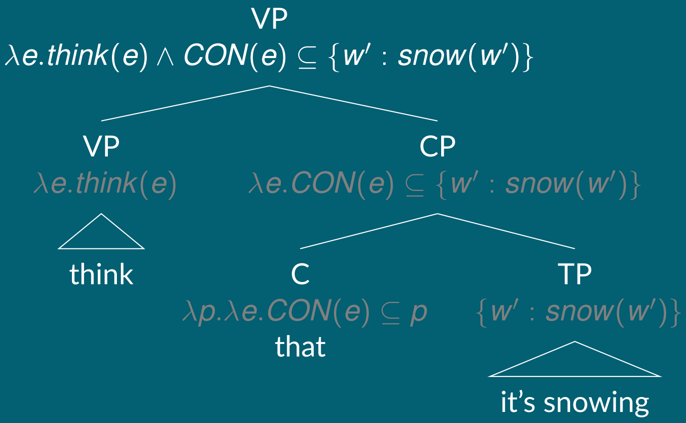

The modification strategy — Elliott (2017), a.o.

Two (among many) competing answers

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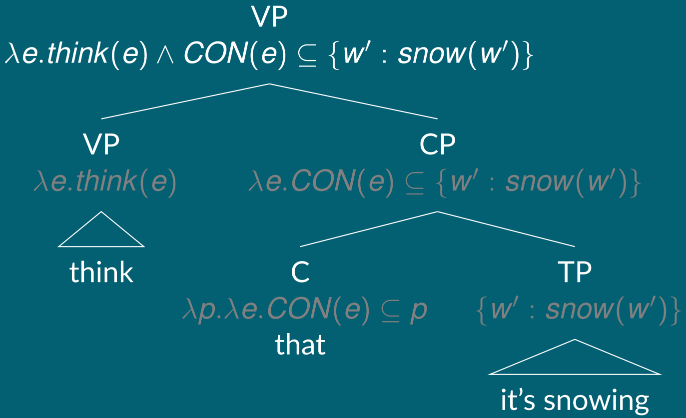
- ✧ Attitude reports' interpretive properties are mostly decided by complementizers.



$\exists e : think(e) \wedge CON(e) \subseteq p \Rightarrow BEL(holder(e)) \subseteq p$